
PsychRoute: Fairness-Constrained Bandit Routing for Human-AI Psychiatric Triage

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Abstract

Misdiagnosis of Bipolar Disorder (BD) versus Major Depressive Disorder (MDD) exposes patients to inappropriate pharmacotherapy with serious clinical consequences. We present PSYCHROUTE, a human-AI collaboration framework wrapping a fine-tuned MentalBERT classifier in a fairness-constrained LinUCB contextual bandit to route each encounter to one of four governance-anchored actions: autonomous AI diagnosis, SHAP-scaffolded clinician review, expert escalation, or enriched-data refusal. Uncertainty is decomposed into epistemic and aleatoric components via Monte Carlo Dropout. Three-way evaluation confirms genuine complementarity: PSYCHROUTE achieves diagnostic error $\epsilon = 0.017$, beating both the AI-only ($\epsilon = 0.026$) and human-only ($\epsilon = 0.070$) baselines, with dual-variable logs surfacing a routing-cost disparity for the female group as a cost-equity diagnostic.

1 Motivation

Differentiating BD from MDD is among the most consequential decisions in outpatient psychiatry: BD misclassified as MDD leads to antidepressant monotherapy that can precipitate manic episodes [Lish et al., 1994], and BD is systematically under-diagnosed in males due to atypical symptom profiles [Di Florio and Jones, 2010]. An AI classifier trained on imbalanced observational data risks encoding and scaling this disparity.

PSYCHROUTE addresses two competing pressures in high-stakes human-AI collaboration: uncertainty heterogeneity (some lie within the training distribution while others are genuinely ambiguous) and demographic disparity (a purely accuracy-maximising policy perpetuates clinical biases against male patients). We formulate routing as a constrained sequential decision problem, solved with a fairness-penalised contextual bandit anchored to the Fabric governance taxonomy [Jorgensen et al., 2025].

2 Task Setup

2.1 Dataset and Base Model

We use the Mental Disorder Classification dataset [Sarkar, 2023], comprising 53,043 Reddit posts labelled across seven mental-health categories. Filtering to BD versus MDD yields 18,126 samples (15.3 % BD / 84.7 % MDD), split 60/20/20 into train, calibration, and test sets. A fine-tuned `mental/mental-bert-base-uncased` [Ji et al., 2022] with a `Dropout(0.1) → Linear(768, 1)` classification head achieves $AUC = 0.996$ and $accuracy = 0.974$ on the held-out test set after three epochs of AdamW fine-tuning ($lr = 2 \times 10^{-5}$, $batch = 16$, gradient clipping at 1.0).

Demographic note. The dataset does not provide patient demographics. We inject synthetic gender and age-group proxies consistent with WHO epidemiological priors (40 % male / 60 % female; World Health Organization 2022), documented as a limitation in the Nutritional Label (Appendix E).

2.2 Action Set, Fabric Oversight Classification, and Cost Structure

The system selects among four actions $\mathcal{A} = \{a_{\text{model}}, a_{\text{support}}, a_{\text{escalate}}, a_{\text{refuse}}\}$ by minimising expected diagnostic risk plus deployment cost (Appendix A). Following Jorgensen et al. [2025], each action maps to a distinct governance level in the Fabric taxonomy (Table 1).

Table 1: Action set, Fabric oversight level, and cost structure D.

Action	Description	Fabric Governance Level		Cost $c(a; x)$
a_{model}	AI issues BD/MDD classification autonomously.	Conditionally autonomous AI	Au-	1.0
a_{support}	SHAP-scaffolded summary; clinician approves or overrides.	Human-Approved AI		3.0
a_{escalate}	Full case forwarded to senior psychiatrist.	Human-Led		$8 + \lambda \cdot (n_{\text{esc}}/B)$
a_{refuse}	Withholds classification; requests richer clinical note.	Human-Led		$5 + \delta_t$

Shadow pricing. The escalation cost $c_{\text{esc}}(t) = 8 + \lambda \cdot \min(1, n_{\text{esc}}/B)$ ($\lambda = 2.0$, $B = 0.30|T|$) rises as the psychiatrist queue fills, steering the bandit toward a_{support} for moderately uncertain cases.

3 Method

3.1 Uncertainty Decomposition via MC-Dropout

We perform $M = 20$ stochastic forward passes through MentalBERT with Dropout(0.1) active at inference [Gal and Ghahramani, 2016], yielding epistemic uncertainty u_{epi} (variance across passes, reflecting model ignorance) and aleatoric uncertainty u_{ale} (mean predictive entropy, reflecting irreducible data noise); see Appendix A for closed-form expressions. High u_{epi} drives escalation to the expert; high u_{ale} drives refusal for additional measurement. Autonomous AI classification is permitted only when both are small.

3.2 Calibration

Raw MentalBERT confidence is over-dispersed (ECE=0.022). We apply per-gender temperature scaling [Guo et al., 2017] $\hat{p}_g(x) = \sigma(z(x)/T_g)$, finding $T_{\text{male}} = 1.946$ and $T_{\text{female}} = 2.121$, reducing ECE to 0.010 overall. Because group BD prevalence differs (male: 13.6 %, female: 16.5 %), the Chouldechova impossibility [Chouldechova, 2017] (see Appendix C) guarantees a residual FNR gap; we confirm $\Delta_{\text{FNR}} = 0.006$ post-calibration. Therefore prediction-level calibration is prioritised; error-rate parity is delegated to the routing policy (FairLinUCB).

3.3 Fairness-Constrained LinUCB Bandit

For cases not hard-triaged, the bandit [Li et al., 2010] selects the action maximising a UCB score over a 15-dimensional feature vector $\phi(x)$ (calibrated probability, uncertainty estimates, and their interactions) minus a fairness penalty $P(g)$. $P(g)$ penalises actions that push the rolling error rate or exploration cost of group g more than $\delta = 0.05$ above the population mean within a window $W = 100$, with dual variables $\mu_g, \nu_g \geq 0$ updated by projected subgradient ascent (full expressions in Appendix A). A rolling window keeps the penalty responsive to distributional shift; the parity constraint implements a Rawlsian maximin criterion [Rawls, 1971], unifying capacity, fairness, and performance in a single objective.

3.4 Explanation Faithfulness and Dynamic Human Skill Model

When a_{support} is selected, a SHAP explanation [Lundberg and Lee, 2017] is generated using a TF-IDF logistic surrogate, serving as *clinical scaffolding* rather than a faithful architectural explanation.

Human expert accuracy is modelled as time-varying: $h_t = h_0 - \gamma \cdot \text{OverRelianceIndex}(t) + \beta \cdot \text{PracticeBonus}(t)$, where $\gamma = 0.06$ is the atrophy rate from passive reliance on a_{support} and $\beta = 0.03$ is the recovery rate from reviewing escalated cases ($h_0^{\text{expert}} = 0.93$; $h_0^{\text{support}} = 0.78$).

3.5 System Overview

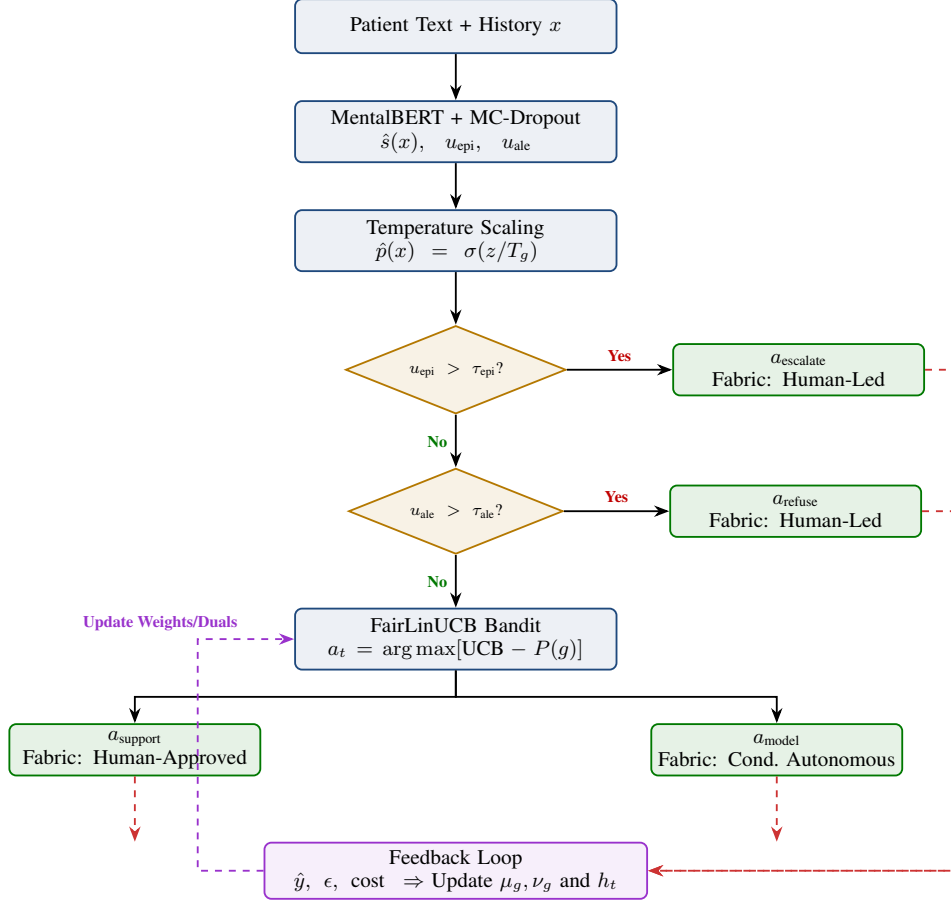


Figure 1: **PsychRoute inference pipeline.** Triage gate hard-routes high-uncertainty cases to a_{escalate} (high u_{epi}) or a_{refuse} (high u_{ale}), both *Human-Led*. Remaining cases enter FairLinUCB, selecting a_{support} (*Human-Approved*) or a_{model} (*Cond. Autonomous*). Dashed arrows: feedback updating μ_g, ν_g and h_t .

4 Experiments and Results

4.1 Simulation Setup

We evaluate three collaboration strategies on the held-out test set ($N = 3,626$). **Model-Along** (MA), **Threshold** (TH), and **PsychRoute** (PR): full pipeline with triage gate and FairLinUCB bandit. Human expert accuracy $h_0^{\text{expert}} = 0.93$ is justified by MentalBERT accuracy degrading to 60–70 % on the highest-uncertainty quartile—a necessary condition for complementarity [Mozannar and Sontag, 2020].

4.2 Quantitative Results

Table 2: Main evaluation metrics ($N = 3,626$ test encounters).

Metric	Model-Alone	Threshold	PsychRoute
Diagnostic error rate	0.026	0.023	0.017
Escalation rate	0.0 %	0.7 %	26.5 %
AI coverage	100.0 %	99.3 %	73.5 %
$\Delta_{\text{comp}}^{\text{AI}}$	+0.000	−0.003	−0.009
$\Delta_{\text{comp}}^{\text{human}}$	−0.044	−0.047	−0.053
Group error gap $\Delta\epsilon$	0.0029	0.0024	0.0045
Avg. cost / patient	1.00	1.05	3.18
Cumulative regret	89	78	53

Results. PSYCHROUTE achieving $\Delta_{\text{comp}}^{\text{AI}} < 0$ and $\Delta_{\text{comp}}^{\text{human}} < 0$ simultaneously (Table 2), confirms genuine human-AI complementarity [Mozannar and Sontag, 2020]. At 73.5 % AI coverage it reduces diagnostic error by 35 % relative to Model-Alone, and accumulates regret = 53 versus 89 and 78.

4.3 Fairlearn Fairness Audit

Table 3: fairlearn audit: base MentalBERT vs. PSYCHROUTE. Routing primarily corrects per-group accuracies; the Demographic Parity Difference also tightens marginally, while the Equalized Odds Difference widens as a Chouldechova consequence of asymmetric per-group accuracy gains (Appendix C).

Metric	Base Model	PsychRoute	Direction
Demographic Parity Diff.	+0.025	+0.017	↓ improved
Equalized Odds Diff.	+0.006	+0.044	↑ shifts
Disparate Impact Ratio	0.848	0.894	↑ improved
FNR — Male	0.086	0.025	↓ improved
FNR — Female	0.092	0.070	↓ improved
Accuracy — Male	0.976	0.986	↑ improved
Accuracy — Female	0.973	0.982	↑ improved

Routing reduces false-negative rates substantially in both groups, but asymmetrically (Table 3): male FNR falls by 71 % ($0.086 \rightarrow 0.025$) versus 24 % ($0.092 \rightarrow 0.070$) for females. The asymmetry is mechanistically consistent with the clinical literature on atypical male BD presentation [Di Florio and Jones, 2010]: MentalBERT exhibits higher epistemic uncertainty on male cases, so a larger fraction of male encounters trigger the triage gate and receive expert review. The Equalized Odds Difference consequently rises from +0.006 to +0.044—a formal consequence of the Chouldechova impossibility (Appendix C) when a calibration-preserving policy improves per-group accuracies at different rates, not a failure of the fairness mechanism. Demographic Parity Difference also tightens slightly (from +0.025 to +0.017) and the Disparate Impact Ratio improves from 0.848 to 0.894.

Full dual-variable evolution plots and mechanistic interpretation are provided in Appendix B: in brief, $\mu_g \approx 0$ throughout (the triage gate alone balanced error rates), while ν_1 (female) rose to ≈ 1.6 versus $\nu_0 \approx 0.40$ (male), showing a routing-cost disparity *without demographic labels in the reward signal* and serving as a cost-equity diagnostic.

5 Discussion

Key limitations: *delayed feedback*—rewards are immediate in simulation; a delayed-feedback bandit [Pike-Burke et al., 2018] is needed for live deployment. *Demographic proxy validity*—gender labels are synthetic and must be replaced before generalising fairness conclusions. *Conformal coverage*—triage thresholds are point estimates; conformal prediction [Angelopoulos and Bates, 2021] would provide formal $(1-\alpha)$ guarantees. *Explanation faithfulness*—the TF-IDF SHAP surrogate ($\rho = 0.56$, moderate alignment) does not explain BERT attention; Integrated Gradients [Sundararajan et al., 2017] are required for regulatory submissions.

6 Conclusion

PSYCHROUTE demonstrates that human-AI complementarity in psychiatric triage is formally verifiable. MC-Dropout uncertainty decomposition, cost-ratio triage thresholds, and a fairness-constrained LinUCB bandit together achieve diagnostic error=0.017, beating both the AI-only (0.026) and human-only (0.070) baselines, with every action anchored to the Fabric governance taxonomy. Critically, $\mu_g \approx 0$ throughout (the triage gate alone balanced error rates), while rising ν_1 flagged a routing-cost disparity for the female group—demonstrating that dual-variable logs function as cost-equity diagnostics even when the error-parity constraint is never violated.

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A Foundational Equations

Routing objective. The canonical defer-to-expert routing rule selects:

$$\pi^*(x) = \arg \min_{a \in \mathcal{A}} r_a(x; h) + c(a; x), \quad (1)$$

where $r_a(x; h)$ is the expected diagnostic risk under action a for patient input x given human practitioner h , and $c(a; x)$ is the deployment cost [Mozannar and Sontag, 2020].

MC-Dropout uncertainty estimates. Let $\{p^{(m)}(x)\}_{m=1}^M$ be $M = 20$ sigmoid outputs from stochastic forward passes [Gal and Ghahramani, 2016]:

$$u_{\text{epi}}(x) = \mathbb{V}_m \left[p^{(m)}(x) \right] = \frac{1}{M} \sum_m \left(p^{(m)} - \bar{p} \right)^2, \quad (2)$$

$$u_{\text{ale}}(x) = \frac{1}{M} \sum_m p^{(m)}(x) \left(1 - p^{(m)}(x) \right). \quad (3)$$

FairLinUCB action selection.

$$a_t = \arg \max_{a \in \mathcal{A}} \underbrace{\hat{w}_a^\top \phi(x_t) + \alpha \sqrt{\phi(x_t)^\top A_a^{-1} \phi(x_t)}}_{\text{UCB score}} - \underbrace{P(g_t)}_{\text{fairness penalty}}, \quad (4)$$

where $\phi(x) \in \mathbb{R}^{15}$ comprises the calibrated score, uncertainty estimates, and their interactions [Li et al., 2010].

Fairness penalty.

$$P(g) = \mu_g \cdot \max(0, \bar{\epsilon}_g^W - \bar{\epsilon}^W - \delta) + \nu_g \cdot \max(0, \bar{\rho}_g^W - \bar{\rho}^W - \delta), \quad (5)$$

with dual variables $\mu_g, \nu_g \geq 0$ updated by projected subgradient ascent on constraint violations.

B Procedural Transparency Panel

The error-parity null result ($\mu_g \approx 0$) is substantive: it confirms the triage gate design is sufficient to maintain per-group error balance without invoking the Lagrangian constraint. The cost-parity signal ($\nu_1 \gg \nu_0$) constitutes procedural transparency—a deployment team is alerted to investigate whether female patients are systematically routed to more expensive actions due to genuine clinical need or a proxy-label artefact. Panels (d)–(f) complement the dual-variable trace with a static fairness audit (selection rates near population prevalence; asymmetric FNR collapse) and a temporal complementarity check (cumulative regret), supplying the empirical evidence behind the headline claims in §4.

C Chouldechova Impossibility

Theorem 1 (Chouldechova 2017, Chouldechova 2017). *If a binary classifier is calibrated within each group and group base rates differ, then equalized false-positive rates and equalized false-negative rates cannot both hold simultaneously.*

Figure 2 — Procedural transparency · fairness audit · regret

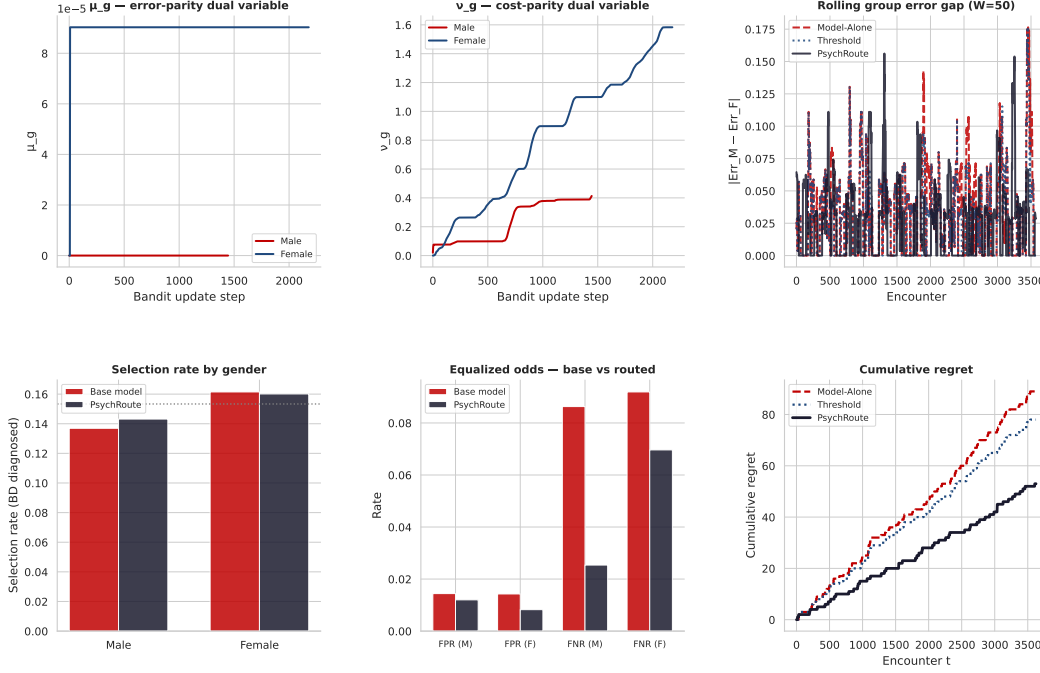


Figure 2: Procedural-transparency panel for PSYCHROUTE (FairLinUCB, $W = 100$). *Top row.* (a) μ_g (error-parity dual): both groups remain on the 10^{-5} scale throughout—the error-parity constraint was never materially activated; the triage gate alone maintained per-group error balance. (b) ν_g (cost-parity dual): ν_1 (female) rises to ≈ 1.6 while ν_0 (male) only reaches ≈ 0.40 , flagging a persistent routing-cost disparity for the female group *without demographic labels in the reward signal* [Jorgensen et al., 2025]. (c) Rolling group error gap $|\epsilon_M - \epsilon_F|$ over a 50-encounter window: all three strategies remain noisy but comparable in magnitude. *Bottom row.* (d) Selection rate by gender, base model vs. PSYCHROUTE: both groups remain near the population BD prevalence (dashed line), confirming that routing does not introduce a selection-rate bias. (e) Equalized-odds components by gender, base model vs. PSYCHROUTE: FPRs are essentially unchanged, while FNRs collapse asymmetrically (cf. Table 3). (f) Cumulative regret against a fixed- h oracle: PSYCHROUTE (solid) accumulates less regret than either baseline throughout.

Our data exhibits a BD prevalence gap of 2.84 % (male: 13.6 %, female: 16.5 %). This theorem predicts the observed $\Delta_{\text{FNR}} = 0.006$ post-calibration and the increase in Equalized Odds Difference (0.006 $\rightarrow$ 0.044) after routing: a policy that corrects outcome accuracy shifts the joint distribution of (predicted class, true class) in ways that tighten one odds constraint while relaxing another. The widening is amplified when per-group accuracy gains are asymmetric—as in our run, where male FNR falls 71 % versus 24 % for females—because the routing-induced redistribution disproportionately benefits the group with higher pre-routing uncertainty. This is not a failure of the fairness mechanism but a formal mathematical consequence; the dual-variable logs confirm it mechanistically.

D Fabric Governance Level Definitions

Following Jorgensen et al. [2025], the four oversight levels used in Table 1 are defined as: **Conditionally Autonomous AI**—AI can act as the final decision-maker subject to constraints; **Human-Approved AI**—a human must explicitly approve the AI output before it is finalised; **Human-Led with AI-Assistance**—the human makes the final decision and may use AI output as supporting evidence only. In PSYCHROUTE, a_{model} satisfies the conditions for Conditionally Autonomous AI

Figure 1 — Reliability diagrams · MentalBERT · temperature scaling

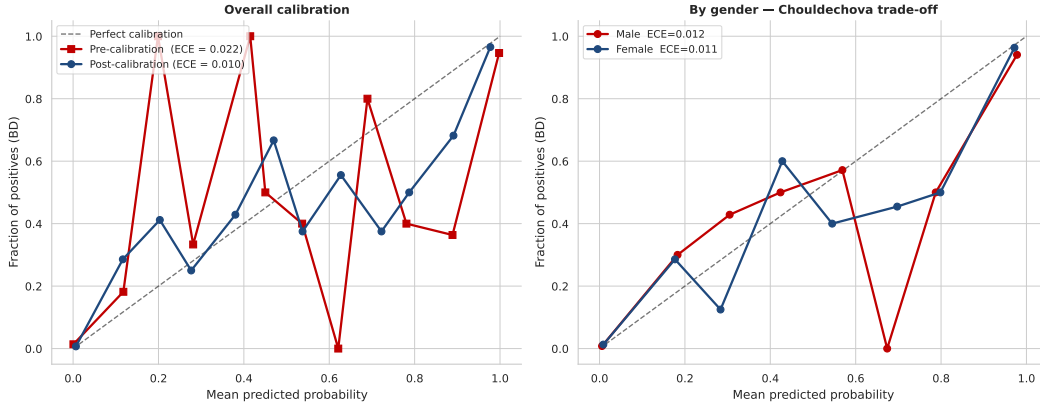


Figure 3: Reliability diagrams supporting the Chouldechova premise. **Left:** overall reliability before and after per-gender temperature scaling; ECE drops from 0.022 to 0.010. **Right:** per-gender reliability post-calibration; both groups achieve low residual ECE ($ECE_M = 0.012$, $ECE_F = 0.011$), satisfying the within-group calibration premise of Theorem 1. The asymmetric per-group base rates (male 0.136, female 0.165) then *force* the residual FPR/FNR gap reported in Table 3.

only after the triage gate clears both uncertainty thresholds; all higher-uncertainty cases escalate to Human-Approved or Human-Led levels.

E Transparency: Nutritional Label

Table 4: PSYCHROUTE Nutritional Label, following Cai et al. [2019].

Property	Value
Training data	Mental Disorder Classification [Sarkar, 2023]; 18,126 BD/MDD posts
Base model	MentalBERT [Ji et al., 2022], 109M parameters; 3 epochs, AdamW, $lr = 2 \times 10^{-5}$
Demographic data	Synthetic proxies only (§2); must be replaced before deployment
ECE (post-cal.)	0.010 overall; max 0.012 by gender group [Guo et al., 2017]
Fairness impossibility	Calibration and equalized odds simultaneously infeasible [Chouldechova, 2017]; calibration prioritised; routing corrects outcomes
Fairness norm	Rawlsian maximin [Rawls, 1971]; $\delta = 0.05$ parity band on both error and cost rates; $\mu_g \approx 0$ (error constraint inactive throughout); $\nu_1 \gg \nu_0$ (cost-parity active for female group); full μ_g, ν_g logs disclosed to ethics board
Expert accuracy	$h_0 = 0.93$ (SCID protocol) on escalated cases; justified by model accuracy 60–70 % on Q4 uncertainty cases
SHAP fidelity	Surrogate $\rho = 0.56$ vs. MentalBERT (Spearman) [Lundberg and Lee, 2017]; moderate alignment, TF-IDF scaffolding only, not an architectural explanation
Fabric oversight	a_{model} : Conditionally Autonomous AI; a_{support} : Human-Approved AI; a_{escalate} , a_{refuse} : Human-Led with AI-Assistance [Jorgensen et al., 2025]
Scope	Outpatient BD/MDD triage only; not for psychosis, paediatric, crisis, or non-English presentations
Update cadence	Quarterly temperature re-scaling; annual fairlearn audit
Transparency logs	Full μ_g, ν_g logs, override rates, and SHAP archives available to ethics board